

Ask Me Anything (AMA) Session

IndoML Datathon 2026

Agenda

- ❑ Introduction
- ❑ Evaluation and Tasks Details
- ❑ Submission Guidelines
- ❑ Q&A Session

Introduction: Datathon Goal

Empowering the next generation of AI innovators



Introduction: Datathon Goal

Empowering the next generation of AI innovators



Prof. Kambhampati, Arizona State Univ

Reasoning and planning abilities of large language models



Prof. Sriraam Natarajan (UT Dallas)

Human-Allied Learning of Tractable Probabilistic Models



Prof. Ravindran Balaraman (IIT Madras)

Learning AI safety from unsafe examples

Introduction: Datathon Goal

Empowering the next generation of AI innovators



Organizers' Details

Datathon Chairs



Dr. Subhajit Datta

Heritage Institute of
Technology



Dr. Mahesh Mohan

IIT Kharagpur



**Dr. Prasanta Kumar
Ghosh**

IISc Bangalore



Dr. Debopriyo Banerjee

Inception - G42

Organizers' Details

</> Technical Volunteers



Shivay Vadhera

IIT Bombay



Nihar Desai

ARTPARK, IISc



Pavan Kumar J

ARTPARK, IISc



Sujith P

ARTPARK, IISc



Shubhadip Nag

Walmart

Organizers' Details

Competition Host (Codabench)



Shubhadip Nag

Walmart

Participation Statistics

Team Registration count: 275+

Teams submitted their prediction in track 1: 45

Teams submitted their prediction in track 2: 15

Tracks 1 and 2

Participants may enter
either **track independently** or **both**

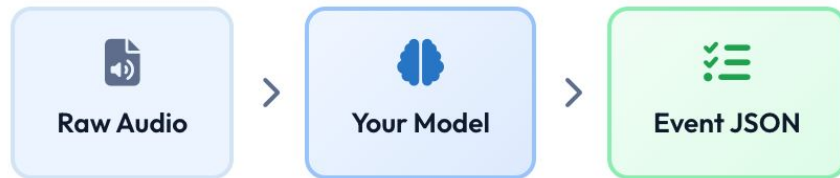
Tracks 1 and 2

01

Detection

Detect Events

Detect noise events in Indic speech recordings with precise onset/offset timestamps. Effectively utilise data annotated at different levels.



```
{onset: 1.24, offset: 3.81}, {onset: 4.31, offset: 4.71}, {onset: 5.04, offset: 5.41}
```

EVALUATION METRICS:

F1

Dice

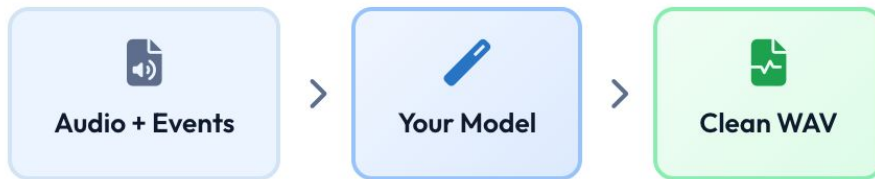
Tracks 1 and 2

Event timeline (onset/offset pairs) from Track 1 is passed as conditioning input to Track 2 — guiding the model on where to suppress noise

02 Removal

Suppress & Preserve

Suppress the detected noise events while preserving the underlying speech signal - output clean, intelligible audio.



16 kHz mono WAV - one per test clip, original filename retained

EVALUATION METRICS:

SI-SDR

Delta WER

PESQ (top-5)

Evaluation Process



Submission

Track 1: Submit a JSON file with onset/offset events per clip.

Track 2: Submit cleaned 16 kHz mono WAV files.



Evaluation

Automated scoring on held-out test clips. Track 1: F1 + Dice. Track 2: SI-SDR + Δ WER, then PESQ for top-5.



Leaderboard

Live rankings published after each submission window. Final standings adjusted by expert Novelty Score for top-5 entries.



Awards

Top teams invited to present at IndoML 2026 and receive cash prizes. Code release required for prize-eligible entries.



Cash Prizes

A total prize pool of **₹2,00,000** —
₹1,00,000 (1 lakh) per track,
awarded across the top teams in
each track.



Present at IndoML 2026

Top teams will be invited to present
their solutions at IndoML 2026, in
front of leading researchers from
academia and industry.



Novelty Award

An expert panel award for the most
original methodological contribution:
new architectures, novel training
regimes, or unsupervised
approaches.

Grand Total Prize Pool: ₹2,00,000

Half-Marathon ₹40K + Final Stage ₹1.5L + Spotlight ₹10K (Track 2)

- Following metric-based ranking, an expert panel applies a novelty factor to the **top-5 entries** in each track to determine the final standings
- Rewards **original contributions**:
 - new **architectures**
 - novel **training regimes**
 - principled **event-conditioning**
 - **semi-supervised approaches** exploiting unlabelled Vaani audio

Prize - Phase 1

🏃 Phase 1 — Half-Marathon Prizes (₹40,000)

Awarded on **20th September 2026** based on leaderboard standings at the midpoint. *Note: The leaderboard will **not** be reset after the half-marathon.*

Category	Per Team	Track 1	Track 2	Total
🥇 🥈 🥉 Ranks 1–3	₹5,000	₹15,000	₹15,000	₹30,000
Ranks 4–8	₹1,000	₹5,000	₹5,000	₹10,000
Half-Marathon Total				₹40,000

Prize - Phase 2

Final Stage Prizes (₹1,50,000)

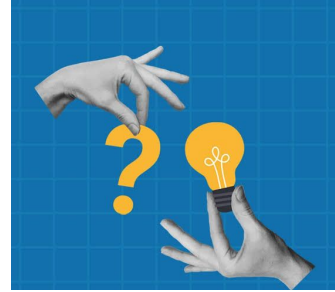
Ranking: Leaderboard (80%) + Novelty (15%) + Report (5%)

Category	Track 1	Track 2	Total
 1st Place	₹35,000	₹35,000	₹70,000
 2nd Place	₹15,000	₹15,000	₹30,000
 3rd Place	₹10,000	₹10,000	₹20,000
4th–8th Place (₹3,000 each)	₹15,000	₹15,000	₹30,000
Final Stage Total			₹1,50,000



Spotlight Solution Award — ₹10,000 (Track 2 only): An additional ₹10,000 reserved exclusively for Track 2 (Noise Event Removal) as a spotlight prize for exceptional solution(s), given that it is the harder problem.

Top-5 entries of **each track** must release **training and inference code** under a permissive open-source licence (MIT / Apache 2.0) and submit a **short (≤ 4 pages) system description**

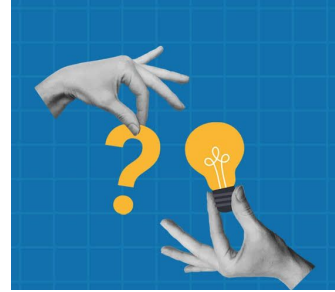


Qn 1. Which Phase is having higher cash prize?

Options: A) Phase 1, B) Phase 2

Qn 2. What is the weightage of novelty?

Options: A) 15%, B) 25%



Qn 1. Which Phase is having higher cash prize?

Options: A) Phase 1, B) Phase 2

Correct Answer: B

Qn 2. What is the weightage of novelty?

Options: A) 15%, B) 25%

Correct Answer: A

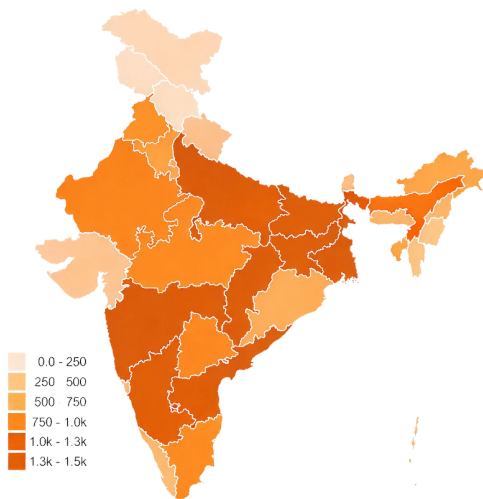


Vaani

**A multi-modal dataset
capturing language
landscape of India**

Project Vaani at a glance

Collected **district by district**, and prompted by **images, not scripts**, so speech is spontaneous and reflects how people actually talk, not how they read.



Recorded hours of speech, by state

31.2K

hours of audio

1,56,534

speakers

105

languages

165

districts

28+3

states & UTs

2,043

hrs transcribed

- **India's First Large-Scale District-Centric Multimodal Multilingual dataset**
- **Represented voices from tribal communities, rural districts, and underrepresented regions**
- **First ever digital corpus for 12 languages**
- **13 Languages not in 2011 census**
- **Collaborated with HF & IndiaAI to host Vaani datasets**
- **Rural skill development**
- **Massive workforce mobilization**

How it's collected and why that's different



IndoML
Indian Symposium on Machine Learning



Vaani

We don't hand people a script. We show them a picture and ask them to talk.



Verified speakers

Local participants registered, verified and onboarded in their own district.



Image-prompted

They describe everyday images in their own words ;spontaneous, not read aloud.



Real-world audio

Recorded the way people really talk; natural accents, switching between languages mid-sentence (like Hindi-English), with real background noise.



Human-in-the-loop QA

Checked by people, not just AI;
Every recording is cut into clips, converted to text, and quality-checked by real people before public release.

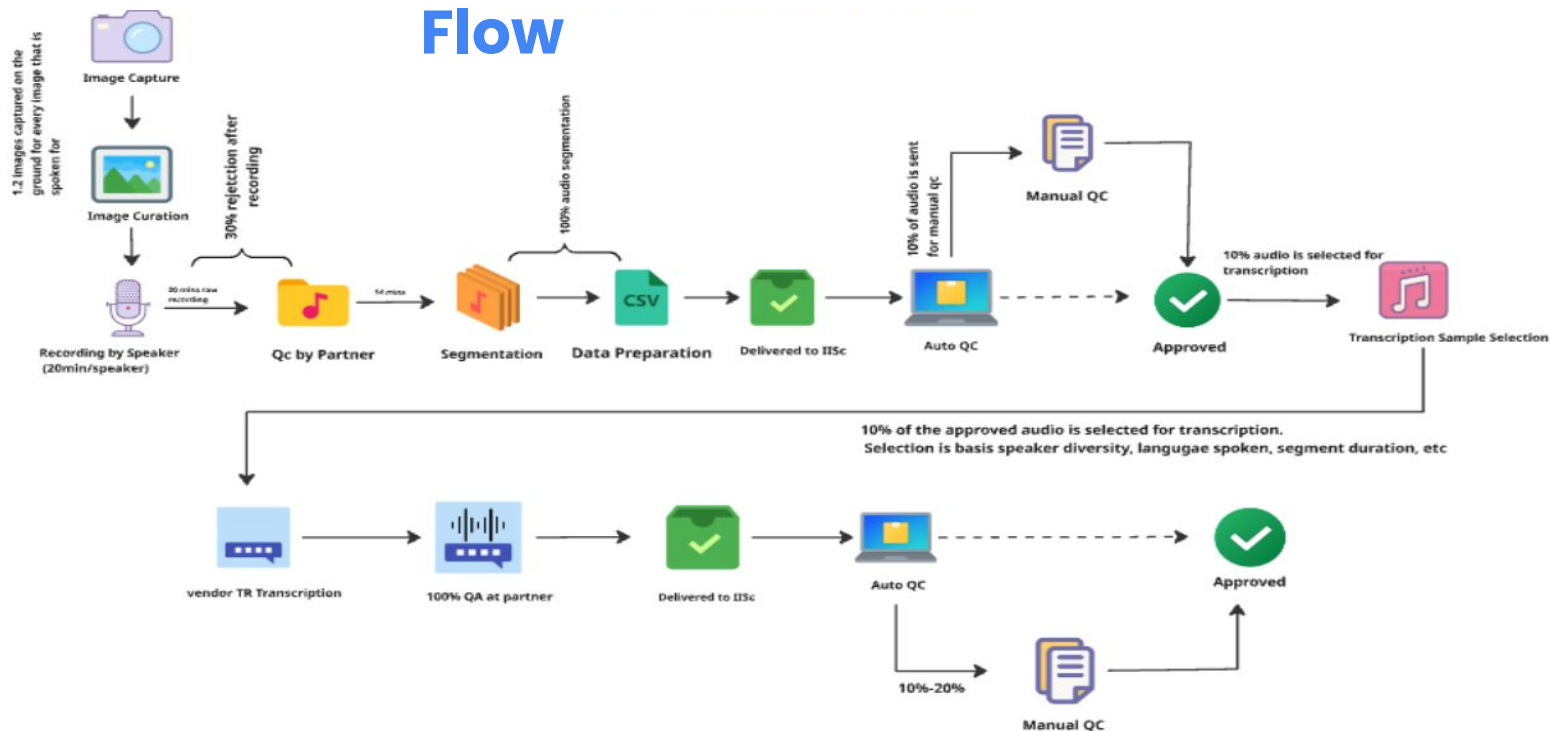
“कोस कोस पर बदले पानी, चार कोस पर वाणी। ”

Geo anchored + Image prompted + Spontaneous speech + native QA = data that reflects how India truly sounds.

Focuses on collecting dialects and languages from remote regions, not just mainstream languages.

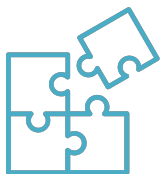


Flow



Why spontaneous,
not read?

Reading a script flattens accent and word choice. A picture prompt draws out spontaneous speech: the dialects, code-mixing and natural phrasing that show how a region truly sounds.



Segmentation Quality

Maintaining consistent segmentation of audio is a complex and ongoing challenge.



Language Discovery

Constant discovery of new languages necessitates continuous script documentation efforts.



Engineering Importance

Robust engineering practices are as critical as machine learning advancements.



Speech Nuances

Balancing mid-silence with natural, spontaneous speech patterns proved difficult.



Dialect Challenges

Current ASR and LLM capabilities struggle with the nuances of regional dialects.



Benchmark Applicability

Western benchmarks like WER may not accurately reflect performance in diverse linguistic contexts.

Some tribal and low-resource languages we captured

LANGUAGE	DISTRICTS	AUDIO HRS
Chakma	3	486
Garó	3	471
Nagamese	4	275
Manipuri	3	207
Mizo	1	202
Wancho	2	121

LANGUAGE	DISTRICTS	AUDIO HRS
Karbi	1	86
Kokborok	4	44
Lotha	2	22
Angami	2	10
Idu Mishmi	1	10
Sumi	3	10

13

languages captured that are
not in the 2011 Census

12

got their **first-ever digital corpus** through Vaani

...and 30+ more tongues from
Arunachal, Nagaland & the wider
Northeast.



The VAANI Noise Event Timestamp Dataset

What This Data **Unlocks**

Because every noise event carries an exact timestamp, three new kinds of experiments become possible



Error attribution, not just detection

WITHOUT SPAN TIMESTAMPS

Clip-level tags: “noise present” — which word broke, and why, stays a mystery

WITH VAANI TIMESTAMPS

VAANI timestamps: pinpoint the exact event — e.g. (dog_bark) 12.40–14.05 s
— behind a specific ASR error



In-span vs. out-of-span evaluation

WITHOUT SPAN TIMESTAMPS

Clip-level corpora: one blended score for the whole clip

WITH VAANI TIMESTAMPS

VAANI timestamps: score suppression inside the noisy span separately
from collateral damage outside it



Span-supervised training signal

WITHOUT SPAN TIMESTAMPS

Weak, clip-level labels: models learn only “noisy vs. not”

WITH VAANI TIMESTAMPS

VAANI timestamps: loss can be weighted exactly where the noise is —
sharper detection, more targeted enhancement

The VAANI Noise Event Timestamp Dataset



72,756

segments



122.17 h

audio



106,892

noise events



58

languages



30 / 162

states / districts



38,541

speakers

Verified (gold)

11,111

segs · 21.85 h

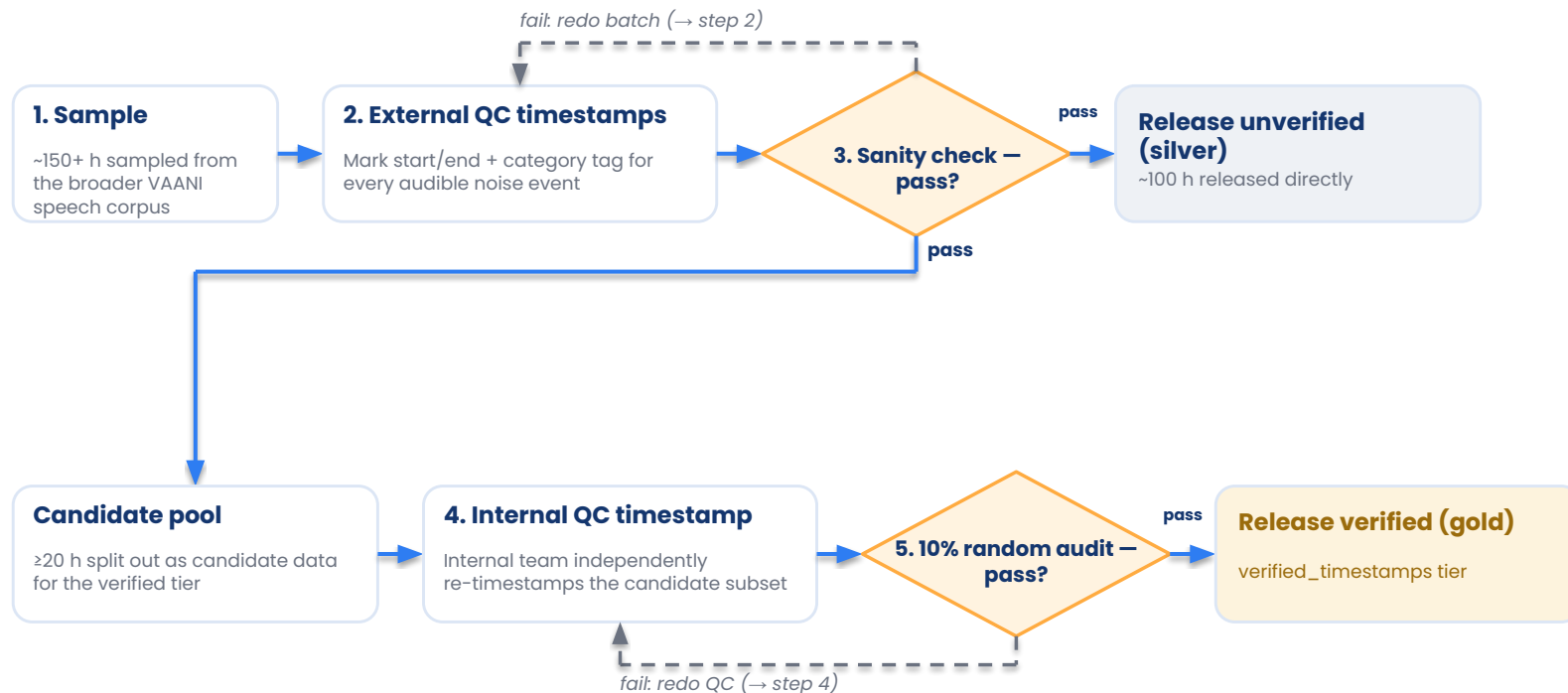
Unverified (silver)

61,645

segs · 100.32 h

Annotation & QC Pipeline

How raw freelancer timestamps become the unverified (silver) and verified (gold) releases



Noise Taxonomy & Quality Control

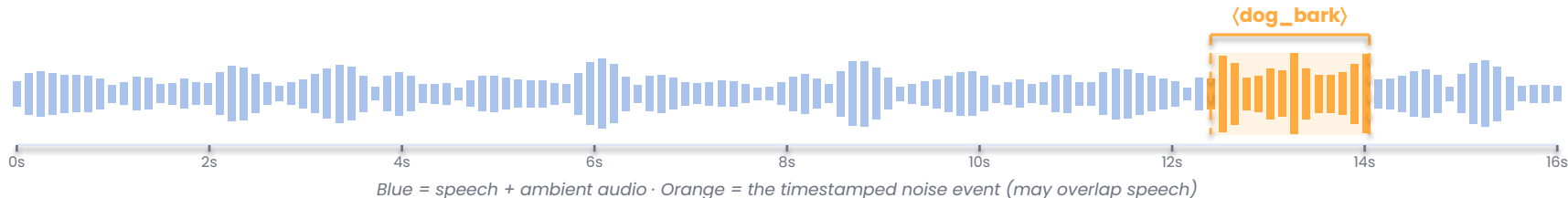
Seven-class noise taxonomy

Category	% segs	events	mean
Non-speech human	37.8%	37,739	0.4 s
Animal	31.3%	24,601	3.3 s
Vehicle / traffic	24.9%	20,603	2.2 s
Baby / child	16.1%	12,376	3.1 s
Singing / music	9.5%	6,978	5.0 s
Phone / signal / alarm	4.9%	3,683	1.8 s
Appliance / machine	1.3%	912	6.1 s

Categories may co-occur ("% segs" need not sum to 100%); "mean" is mean event duration.

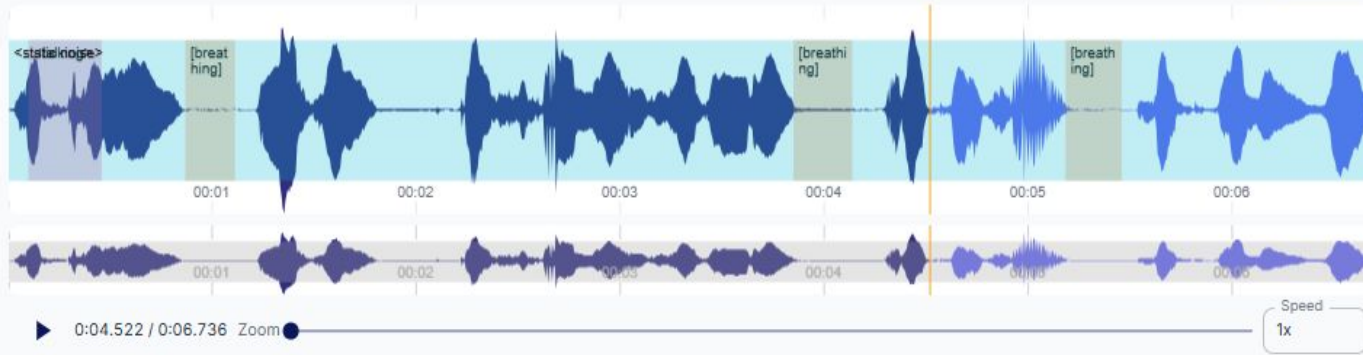
Example: an annotated segment

{category: Animal, tag: <dog_bark>, start: 12.40 s, end: 14.05 s}



Annotate Audio

File: IISc_VaaniProject_M_MadhyaPr... State: MadhyaPradesh District: Katni Language: Hindi

[View Image](#)


#	START (S)	END (S)	DURATION	TAGS
1	0 0:00.000	6.73 0:06.730	0:06.730	<static noise>
2	0.101 0:00.101	0.458 0:00.458	0:00.357	<talking>
3	0.872 0:00.872	1.116 0:01.116	0:00.244	[breathing]
4	3.85	4.141	0:00.291	[breathing]

Original Transcription

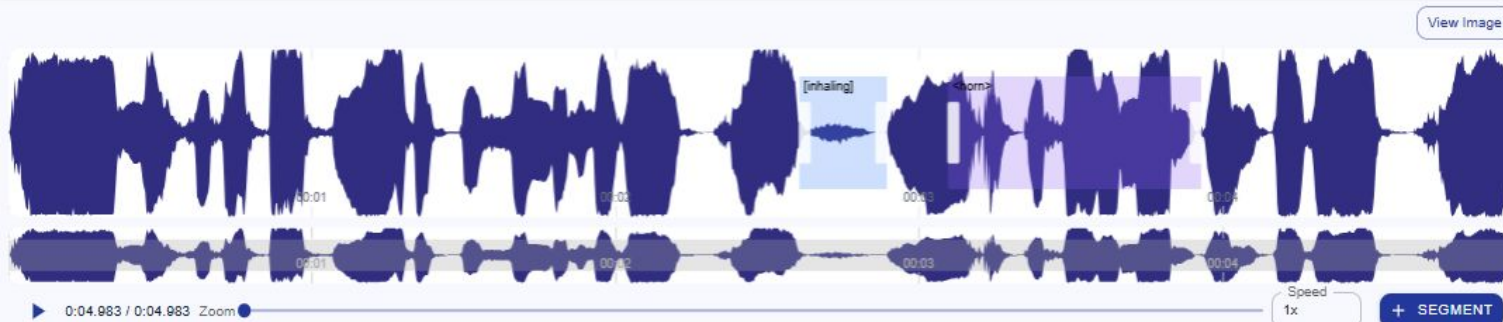
(00:00.000 - 00:06.730) <noise> चित्र
 (00:00.101 - 00:00.458) <talking>
 </talking> में मुझे कुछ सामान वगैरा
 (00:03.753 - 00:04.094) [breathing]
 <PAUSE> रखा सा दिख रहा है। </noise>
 (00:00.980 - 00:01.820)
 [autoGeneratedTag] (00:01.860 -
 00:03.280) [autoGeneratedTag]

Update Transcription

<static noise> चित्र <talking> </talking>
 में [breathing] मुझे कुछ सामान वगैरा
 [breathing] <PAUSE> रखा सा
 [breathing] दिख रहा है। </static noise>

Annotate Audio

FILE: Jharkhand_Jamtara_Sina56988_11... STATE: Jharkhand DIST: Jamtara LANG: Bengali ASSIGNEE: Ankit Chamoli Jharkhand_Jamtara_... Bengali



View Image

Original Transcription

আ মসজিদের পাশেও আরেকটি
বিল্ডিং {building} দেখা যাচ্ছে।
(00:02.574 - 00:02.977) [inhaling]
একটি (00:02.603 - 00:03.434)
<horn> </horn> বড় ল্যাম্প {lamp}
পোস্ট {post} দেখা যাচ্ছে।

Update Transcription

Auto-fill

আ মসজিদের পাশেও আরেকটি
বিল্ডিং {building} দেখা যাচ্ছে।
[inhaling] একটি <horn> </horn> বড়
ল্যাম্প {lamp} পোস্ট {post} দেখা
যাচ্ছে।

#	Start (s)	End (s)	Duration	Tags		
1	2.603 0:02.603	2.896 0:02.896	0:00.293	Select or type tag [inhaling]	▶	🗑️
2	3.096 0:03.096	3.927 0:03.927	0:00.831	Select or type tag <horn>	▶	🗑️

CANCEL

SAVE FOR LATER & NEXT

SUBMIT & NEXT

SUBMIT & CLOSE ▶



From Data to Experiments

One tagged segment, two experiments — pipeline, architecture, and eval formula



Competition Website

IndoML

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Sponsorship

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Attend

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Register now

Registration

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Timeline

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Register Now

Both tracks are now LIVE on CodeBench! [Track 1](#) [Track 2](#) [Join Discord](#)

IndoML

Launched this Independence Day — 15th August 2026

Vaani

Join our Discord

COMPETITION STARTED

15th August — 80th Independence Day

PHASE 1 — HALF-MARATHON

PHASE 2 — FINAL DEADLINE

21st 10th 4th 5th

26th 14th 4th 5th

Independence Day Challenge - IndoML 2026

Datathon@IndoML

2026

Cancel

Complete Now

CELEBRATING INDIA'S LINGUISTIC DIVERSITY THROUGH ML

Noise event detection and removal in real-world Indian speech — be part of India's ML community. Build something meaningful for Bharat. [Go](#)

Open to everyone based in India — students, researchers & industry professionals

₹2,00,000

Prize Pool

Travel Grants

Presented by IndoML

Register Now

Learn More

Track 1 on CodeBench

Track 2 on CodeBench

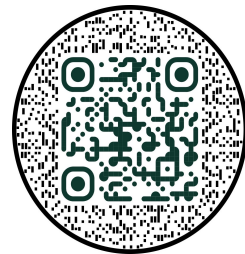
Training Outline

About the Datathon

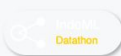
Welcome to **Datathon@IndoML 2026** — a research-oriented data science competition held in conjunction with **IndoML 2026**. Building on the success of previous editions, this year's datathon challenges participants to tackle **noise event detection and removal in real-world Indian speech** — a critical problem for Indian, robust Automatic Speech Recognition across Indian languages.

The competition is organized into two tightly coupled tracks: **Track 1 (Detection)** - detect noise events with precise timestamps, effectively offering data enriched at different levels - and **Track 2 (Removal)** - suppress detected events while preserving the underlying speech. The dataset is a curated subset of the **Noise corpus**, consisting of ~15k hours of real-world Indian audio with three levels of annotation quality.

Top-performing teams will be invited to attend **IndoML 2026** and present their solutions to leading researchers and professionals from academia and industry. These teams will also receive exciting **cash prizes**.



Competition @ CodaBench

Datathon

DATATHON@INDOML 2026 —
TRACK 1: NOISE EVENT
DETECTION

148 PARTICIPANTS
489 SUBMISSIONS

ORGANIZED BY: Nagu_iitkgp
CURRENT PHASE ENDS: October 15, 2026 At 12:00 PM GMT+5:30
CURRENT SERVER TIME: September 8, 2026 At 7:12 PM GMT+5:30
Docker image: codaib/codaib-legacy:py312

Get Started

Phases

My Submissions

Results

Forum

Overview

Data

Evaluation

Terms

Datathon@IndoML 2026 — Track 1: Noise Event Detection

TL;DR

Task	Detect noise events with precise onset/offset timestamps in real-world Indic speech.
Submit	A ZIP containing a single <code>predictions.jsonl</code> — one JSON object per clip.
Metric	Combined = Event-based F1 ($\pm 20\%$ tolerance) + Segment-level Dice. Max = 2.0.
Data	~150 h training (Gold / Silver / Bronze tiers), 11 h withheld test (7 h natural + 4 h synthetic).
Training Data	ARTPARK-IISc/Vaani-Noise-Event-Dataset on Hugging Face
Test Data	Go to "Get Started" → "Files" and download <code>input_data</code> .
Validation Dataset	In the same "Get Started" → "Files" section, download the validation dataset file.
Deadlines	Phase 1 (Half-Marathon): Sep 20, 2026 · Phase 2 (Final): Oct 15, 2026
Register	Google Form (compulsory, closes 5 Sep) + CodaBench.
Other track	Track 2 — Noise Event Removal.

Datathon

DATATHON@INDOML 2026 —
TRACK 2: NOISE EVENT
REMOVAL

50 PARTICIPANTS
28 SUBMISSIONS

ORGANIZED BY: Nagu_iitkgp
CURRENT PHASE ENDS: October 15, 2026 At 12:00 PM GMT+5:30
CURRENT SERVER TIME: September 8, 2026 At 7:12 PM GMT+5:30
Docker image: nagu_iitkgp/datathon-removal-scoring:latest

Get Started

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My Submissions

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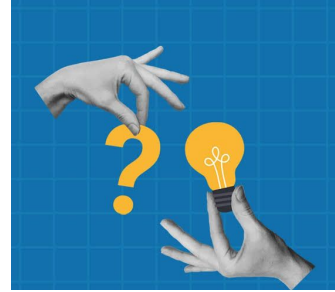
Terms

Datathon@IndoML 2026 — Track 2: Noise Event Removal

TL;DR

Task	Suppress detected noise events in real-world Indic speech while preserving the underlying speech.
Submit	A ZIP with enhanced 16 kHz mono WAVs (one per clip) + a <code>transcripts.jsonl</code> from the mandated ASR.
Metric	Combined = SI-SDR (synthetic subset) + 100 × Δ WER (full test set). A hidden ASR audit checks the transcripts.
Data	~150 h training (Gold / Silver / Bronze tiers), 11 h withheld test (7 h natural + 4 h synthetic).
Training Data	ARTPARK-IISc/Vaani-Noise-Event-Dataset on Hugging Face
Test Data	Go to "Get Started" → "Files" and download <code>input_data</code> .
Validation Dataset	In the same "Get Started" → "Files" section, download the validation dataset file.
ASR Model	ARTPARK-IISc/SraVaani-1.0 (frozen, mandated for Δ WER and transcripts.jsonl)
Deadlines	Phase 1 (Half-Marathon): Sep 20, 2026 · Phase 2 (Final): Oct 15, 2026
Register	Google Form (compulsory, closes 5 Sep) + CodaBench.
Other track	Track 1 — Noise Event Detection.

Platform Walkthrough!!!

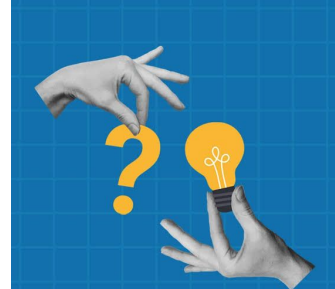


Q1. What is the encoder used in Track-1?

A) WavLM-base+ B) SraVaani FastConformer C) BiGRU D) Others

Q2. The denoiser predicts a _____ for Track-2.

A) Frame timeline B) Magnitude mask C) Clip tag D) Onset/offset



Q1. What is the encoder used in Track-1?

A) WavLM-base+ B) SraVaani FastConformer C) BiGRU D)

Others

Correct: A — WavLM's 50 Hz frames match the 20 ms label grid (BiGRU is other part of the pipeline; FastConformer is Track 2's).

Q2. The denoiser predicts a _____ for Track-2.

A) Frame timeline B) Magnitude mask C) Clip tag D)

Onset/offset

Correct: B — A mask on the STFT, then iSTFT back to audio.